

MONA, United Kingdom

WMO: 033010

Lat: 53.2600N

Lon: 4.3761W

Elev: 203

StdP: 14.59

Time Zone: 0.00 (EUW)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	26.1	29.0	21.8	16.5	29.8	24.3	18.6	31.5	41.7	47.6	37.9	47.6	4.2	60	0.709

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.6	72.8	63.0	69.3	61.5	66.7	60.3	64.6	69.6	63.0	66.9	61.7	64.9	10.2	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.7	85.9	66.1	61.5	82.2	64.4	60.3	78.8	62.9	29.7	69.6	28.6	67.1	27.7	65.0	73.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
34.0	30.0	26.8	DB	21.4	79.4	5.4	4.0	17.5	82.3	14.3	84.6	11.3	86.9	7.3	89.8
			WB	20.6	68.5	5.3	2.7	16.8	70.4	13.7	72.0	10.7	73.5	6.8	75.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.6	41.7	41.3	43.0	46.8	51.4	56.2	59.2	58.4	55.6	51.8	45.6	43.3
	DBStd	7.88	5.00	4.95	4.72	4.52	4.44	3.81	3.66	3.08	3.86	4.75	5.25	5.96
	HDD50	1275	258	246	221	116	32	2	0	0	3	36	150	211
	HDD65	5637	722	665	683	545	421	267	184	204	281	410	583	672
	CDD50	1119	1	1	3	21	76	187	285	261	172	91	16	5
	CDD65	6	0	0	0	0	0	2	4	0	0	0	0	0
	CDH74	61	0	0	0	0	6	20	31	2	2	0	0	0
	CDH80	4	0	0	0	0	0	1	3	0	0	0	0	0
Wind	WSAvg	12.3	13.5	13.6	12.5	11.7	12.0	11.2	10.9	11.9	11.6	12.6	11.9	14.4
Precipitation	PrecAvg	33.3	3.1	2.3	2.4	2.1	2.0	2.1	2.2	2.8	2.9	3.8	4.1	3.7
	PrecMax	41.5	6.3	6.3	6.0	4.1	5.0	5.1	5.0	5.6	6.3	8.2	8.7	10.0
	PrecMin	24.6	0.3	0.3	0.5	0.3	0.4	0.4	0.4	0.4	0.1	0.7	1.1	0.7
	PrecStd	3.9	1.4	1.2	1.1	1.0	1.1	1.1	1.1	1.1	1.5	1.9	1.6	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.3	56.5	63.2	69.4	73.2	77.1	77.3	72.0	71.7	65.9	58.5	55.8
		MCWB	52.3	48.6	52.6	57.6	62.7	63.2	66.3	63.1	64.0	60.6	55.4	53.3
	2%	DB	51.3	51.2	54.6	61.9	66.2	70.4	73.7	68.1	67.5	61.9	56.8	53.7
		MCWB	50.6	48.4	49.8	53.4	58.9	61.7	63.7	61.6	60.7	58.7	55.8	52.8
	5%	DB	49.8	49.6	51.7	57.2	62.2	66.5	69.7	65.7	64.5	60.2	55.2	52.5
		MCWB	48.8	48.3	48.8	50.9	56.0	59.3	61.9	60.4	60.2	58.1	54.2	51.7
	10%	DB	48.9	48.4	49.7	54.2	59.3	63.8	66.8	64.0	62.4	58.9	53.8	51.2
		MCWB	47.8	47.2	47.1	49.4	54.1	58.0	60.9	59.8	58.6	57.0	52.4	50.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.6	51.0	53.3	57.9	64.1	65.9	67.8	64.6	64.8	61.2	57.4	54.5
		MCDB	53.0	53.0	59.5	67.1	70.8	73.5	75.4	67.6	68.8	63.3	57.7	54.7
	2%	WB	50.9	49.7	50.9	54.3	60.3	62.7	65.2	63.0	62.7	60.0	56.0	53.2
		MCDB	51.1	50.6	53.7	60.8	64.9	68.4	71.1	65.8	65.1	61.3	56.4	53.5
	5%	WB	49.4	48.5	49.3	51.9	57.0	60.4	63.3	61.9	60.9	58.9	54.5	51.9
		MCDB	49.7	49.2	51.4	55.4	60.6	64.8	67.6	64.5	63.1	59.8	55.1	52.3
	10%	WB	48.2	47.5	48.0	50.5	55.2	58.8	61.7	61.0	59.7	57.3	52.6	50.5
		MCDB	48.8	48.3	49.4	53.3	58.0	62.6	65.2	63.2	61.6	58.5	53.6	51.1
Mean Daily Temperature Range	5% DB	MDBR	8.8	9.6	11.5	12.8	12.8	12.9	12.6	11.4	12.1	10.3	9.9	8.5
		MCDBR	7.6	9.5	14.1	18.6	17.8	19.4	18.5	13.9	15.8	10.6	9.4	8.4
		MCWBR	8.5	8.0	10.1	11.2	11.6	12.3	10.9	9.2	10.6	8.2	9.2	9.4
	5% WB	MCDBR	7.8	8.3	12.5	15.5	14.7	15.7	15.3	11.6	12.1	9.0	8.7	8.0
		MCWBR	8.8	7.5	9.6	10.1	10.2	10.7	10.0	8.3	9.2	7.9	9.1	9.1
Clear-Sky Solar Irradiance	taub	0.331	0.343	0.368	0.386	0.391	0.383	0.386	0.371	0.364	0.350	0.327	0.320	
	taud	2.310	2.377	2.339	2.317	2.356	2.404	2.401	2.456	2.471	2.450	2.376	2.286	
	Ebn at Noon	200	240	258	268	272	275	272	270	258	235	201	179	
	Edh at Noon	20	25	31	36	36	35	35	31	27	23	18	17	
All-Sky Solar Radiation	RadAvg	206	433	823	1296	1647	1750	1600	1338	955	529	255	153	
	RadStd	19	38	61	91	115	148	138	90	60	47	25	15	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (50 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page